Pseudo Filesystems === proc virtual filesystem

- The `proc` virtual filesystem exists since the beginning of Linux
- It allows
  - The kernel to expose statistics about running processes in the system
  - The user to adjust at runtime various system parameters about process management, memory management, etc.
- The `proc` filesystem is used by many standard user space applications, and they expect it to be mounted in `/proc`
- Applications such as `ps` or `top` would not work without the `proc` filesystem

- Command to mount `proc`: `mount -t proc nodev /proc`
- See filesystems/proc in kernel documentation or `man proc`

proc contents

- One directory for each running process in the system
  - `/proc/<pid>`
  - `cat /proc/3840/cmdline`
  - It contains details about the files opened by the process, the CPU and memory usage, etc.
- `/proc/interrupts`, `/proc/iomem`, `/proc/cpuinfo` contain general device-related information
- `/proc/cmdline` contains the kernel command line
- `/proc/sys` contains many files that can be written to adjust kernel parameters
  - They are called *sysctl*. See admin-guide/sysctl/ in kernel documentation.
  - Example (free the page cache and slab objects): `echo 3 > /proc/sys/vm/drop_caches`

sysfs filesystem

- It allows to represent in user space the vision that the kernel has of the buses, devices and drivers in the system
- It is useful for various user space applications that need to list and query the available hardware, for example `udev` or `mdev` (see later)
- All applications using sysfs expect it to be mounted in the `/sys` directory
- Command to mount `/sys`: `mount -t sysfs nodev /sys`
-
```
\$\_ls /sys/
block bus class dev devices firmware fs kernel module power
```